

Improving selection of forest fuel metrics using synthetic lidar point clouds

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ABSTRACT

Fuel characteristics are key drivers of fire behaviour and an essential consideration in fire management. Terrestrial Lidar Scanners (TLS) have emerged as valuable tools for quantifying fuel structures using three-dimensional point clouds summarised by multiple metrics at the scale of forest stands. Although an impressive array of TLS-derived metrics can be calculated to understand fuel characteristics, we don't know which are most useful or meaningful. Further, many of the metrics currently described were designed for specific forest types, limiting their widespread application. Identifying an optimal subset of useful metrics requires evaluating metrics over scans of many stands with known fuel attributes. However, such extensive data collection would be very costly and time-consuming. To address this challenge, we developed a framework for testing fuel metrics using synthetic point clouds built from subsets of real-world lidar data. We demonstrate this framework in the boreal jack pine forests of northwestern Ontario, Canada. Our approach quickly generates artificial forest stands with predefined and known fuel attributes, by varying four structural parameters: stem density, stem mortality, understory density, and mean tree height. Generated stands represent a broad range of fuel scenarios with pertinent characteristics, such as vertical fuel connectivity or canopy gaps. Metrics are then evaluated on their performance as predictors for fuel scenario clusters. Our results find a parsimonious set of metrics that are likely to be reliable in the fuel characterisation of jack pine stands. This research can also be adapted as a robust method for identifying metrics in any other forest type.

1. Introduction

Fires are a natural and common disturbance in the boreal forest and play a critical role in forest health, regeneration, and ecosystem functioning (Bergeron et al., 2002). However, climate change is increasing fire activity (Barnes et al., 2023; Flannigan et al., 2005) and increasing the risk of negative impacts of fires on forest ecosystems (Baltzer et al., 2021) and communities (Erni et al., 2021). Understanding the factors affecting fire behaviour is important for better adapting fire management strategies to the changing climate.

Fuels are a key component of wildfire behaviour (Burgan and Shasby, 1984; Flannigan et al., 2005) and an essential consideration in fire management (Keane, 2015). Fuels are the live or dead organic material that can combust and burn (Keane, 2015). An important fuel characteristic is fuel structure, which describes the within- and among-stand vertical and horizontal spatial arrangement of combustible materials. Fuel structure influences fire behaviour, including rate of spread, intensity, and the probability of a surface fire transitioning to a crown fire (Keane, 2015). A key structural element is ladder fuels, which

extend upward from the surface layer or downward from the canopy layer, and can facilitate the spread of a surface fire into the canopy fuel layer, becoming a crown fire (Menning and Stephens, 2007). The horizontal connectivity of the canopy fuel layer further enables the crown fire to propagate through the forest stand (Contreras et al., 2012). However, accurately quantifying fuel attributes remains challenging because of their complexity, variability, and three-dimensional structure.

In Canada, fuels are typically represented as one of 16 categorical fuel types based on classified remotely sensed imagery (Forestry Canada Fire Danger Group, 1992). These two-dimensional landscape fuel type maps are represented in raster format and are used for risk assessment, fire behaviour prediction, and simulating expected fire growth (Parisien et al., 2005). Although well suited for operational applications, their scale of representation (i.e., ~30 m) and categorical nature can oversimplify fuel structures and fail to capture important variation within and among stands that is pertinent to fire behaviour and outcomes (Baron et al., 2024; Phelps and Beverly, 2022). Detailed local-scale assessments remain essential for quantifying the fine-scale, three-

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dimensional variation in fuel structure that landscape fuel type maps do not represent.

Fuel structure at the stand scale has traditionally been quantified using manual field measurements built on forest mensuration techniques (Bonham, 2013; Boucher et al., 2024). Established allometric equations are then used to translate within-stand field measurements (typically along a transect) to estimations of stand-level fuel characteristics (e.g. crown bulk density, canopy base height, fuel moisture). Similarly, data from experimental burns in different fuel types are used to translate these direct and derived field measurements into expected fire behaviour (e.g. rate of spread, crowning potential) (Duff et al., 2017; Lambert et al., 2005; Stocks et al., 1989). However, the collection of field vegetation measurements is time-consuming, costly, and can lack precision, resulting in high uncertainty (Duff et al., 2017). A way to augment measurements of stand-level fuel characteristics in a fast and comprehensive manner is to use remote sensing technology (Gale et al., 2021).

In recent years, lidar, or light detection and ranging, has increasingly been used to precisely map three-dimensional forest structures quickly and at a range of scales (Liang et al., 2016; Wallace et al., 2022). Lidar is mainly used for forest surveying and mensuration (Hudak et al., 2009; Wulder et al., 2008), but it has also been used for assessing fuel structures (Ferraz et al., 2009; Wulder et al., 2012). Lidar scanners provide data in the form of a collection of points (a “point cloud”) representing the area scanned. Each point represents the reflection of a laser pulse and has a position in 3D space calculated by the scanner by combining the direction of the pulse and the time to reflection (Béland et al., 2014).

Many studies have used Aerial Lidar Scanners (ALS) or airborne lidar, mounted on aircraft or drones, to quantify forest structure because of their ability to cover large areas cost effectively compared to manual measurements (Hilker et al., 2012). However, ALS data has a lower point density relative to in-stand Terrestrial Lidar Scans (TLS). For this reason, ALS data are typically used to derive relatively simple forest metrics such as stand volume and canopy height (Jakubowski et al., 2013; Nilsson, 1996). On the other hand, TLS provide denser point cloud data over a smaller area (Liang et al., 2016). Because of this, TLS is often preferred over ALS when deriving structural attributes of the forest beyond stand-scale parameters (Paris et al., 2017; Wallace et al., 2022), although there are important trade-offs in spatial extent and data complexity. ALS covers large continuous areas but has limited within-canopy structure coverage, whereas TLS has limited spatial coverage but provides detailed within-canopy structure information (Béland et al., 2019).

Despite being very information rich, extracting meaningful and actionable fuels information from TLS point clouds remains challenging. Although a myriad of metrics has been developed to summarize lidar point clouds, no single set of metrics has been identified that can reliably capture fuel information for all forest types and scanning methods. Metrics are often designed for specific forest types, e.g. eucalyptus forests (Hillman et al., 2021; Wilson et al., 2021), old-field pine-grasslands (Rowell et al., 2020) or Mediterranean shrublands (Alonso-Rego et al., 2020), using a range of TLS sensors and acquisition protocols. When analysing TLS data, it is not clear whether it is possible to correctly understand fuel characteristics when reusing a metric defined for another forest type, for another sensor, or for another scanning configuration. It remains unclear whether metrics developed for one ecosystem or sampling approach will perform well in another. More importantly, we have a limited understanding of the extent to which different metrics capture similar or complementary aspects of forest structure. Comparing metric performance can help to identify metrics that perform consistently across forest condition and quantify different aspects of fuel structure.

Lidar-derived fuel metrics aim to capture stand attributes relevant to expected fire behaviour, such as fuel compactness, type, density and within-stand arrangement. Previous terrestrial lidar-based studies on fuels and associated fire hazard have used height distribution attributes

such as mean, standard deviation, variance, or skewness (Alonso-Rego et al., 2020; Jakubowski et al., 2013; Singh et al., 2018). Various types of volumetric information have also been explored, such as occupied volume or cover (Hudak et al., 2021; Rowell et al., 2020; Wallace et al., 2022). Other studies have developed approaches to recreate traditional manual fuel measurements such as canopy base height or canopy bulk density (Engelstad et al., 2019; Hillman et al., 2021; Wilson et al., 2021). In addition, novel structural metrics have been defined to take advantage of information challenging to acquire without the precision and scale of TLS data. Recent examples include the “porosity” metric, which estimates fuel compactness (Rowell et al., 2020), the stand structure metric “Effective Number of Layers” (Ehbrecht et al., 2016) and quantifying visual occlusion using “isovists” (Batchelor et al., 2023).

Fuel data must be accurate and reliable, given the operational decisions that can depend on them (Duff et al., 2017). Similarly, selected metrics must accurately capture fuel structure and meaningfully distinguish between different stands. Unfortunately, directly evaluating whether a set of metrics accurately captures a real-world stand's fuel structure necessitates extensive valid reference data. The manual field measurements required to establish this information for testing would necessitate a time-consuming, expensive, and potentially error-prone data acquisition process. A well-established alternative that reconciles the need for rigorous testing while avoiding impractical real-world data acquisition is to use simulated data (Zurell et al., 2010) that captures a range of stand conditions, allowing for the evaluation of various metrics and the selection of a subset that most accurately captures variation in fuel structures.

There are two general approaches to simulating lidar point clouds: the first is the simulation of a lidar sensor's laser pulses in a virtual 3D environment. Tools such as Helios++ (Winiwarter et al., 2022) can simulate many types of sensors, including those of ALS and TLS. Applications of this simulation are the creation of datasets for deep learning models for tree-segmentation (Bryson et al., 2024) or completing point clouds that have been occluded (Bornand et al., 2024). The second approach is to construct synthetic point clouds through the combination of point cloud objects that have been segmented from real-world data. An example of this approach is Comesaña-Cebal et al., 2024's synthetic multispectral terrestrial lidar dataset for analysing the fire response of different tree species. In this study, we adopt this latter approach and use real-world segmented data as objects that are combined to form the final simulated point clouds of jack pine (*Pinus banksiana* Lamb.) stands. This approach is less time-consuming and computationally demanding, assuming real-world data is available, as it forgoes the challenges of parameterizing, creating, and simulating a realistic 3D environment for a virtual sensor to scan.

Jack pine (*Pinus banksiana* Lamb.) is one of the most important commercial tree in Canada's boreal forest (Law and Valade, 1994). Jack pine stands are well-adapted to fires, as their serotinous cones release seeds under high heat, which promotes stand regeneration even after high intensity fires (Gauthier et al., 1996). The highest intensity fires in Canada are typically observed in jack pine stands (de Groot et al., 2004). Understanding fuels in jack pine stands is critical to predict fire risk, behaviour and severity in Canada.

In this study, we present a novel framework for generating synthetic forest stands from TLS-derived point cloud objects. The resulting point clouds enable direct comparisons of TLS metrics under known structural conditions. We demonstrate this framework using a range of simulated boreal jack pine stands, with which we evaluate twelve metrics. The metrics we compare include: understory height (UH), fuel strata gap (FSG), canopy base height (CBH) and canopy height (CH) metrics defined in Wilson et al., 2021, the canopy ratio (CR) metric from Maeda et al. (2022), the ¹D Effective Number of Layers (ENL), ²D ENL, and ²D ENL CV metrics from Aalto et al. (2023), the scan-wide ¹D ENL, and scan-wide ²D ENL metrics from Ehbrecht et al. (2016), Cooper, 2022's Total Lacunarity metric and Scott et al., 2022's lacunarity-derived

Heterogeneity Index. These metrics were chosen because they cover a range of philosophies: from tangible metrics (CH, CBH, FSG, UH, CR) that are easy to understand and recreate classic field measurements (e.g. CH, CBH in Van Wagner, 1977; FSG in Cruz et al., 2004) to abstract metrics (ENL metrics and 3D lacunarity metrics) that are more conceptual and take advantage of TLS's data output to capture novel fuels information. Better understanding of which metrics are most effective at distinguishing stand and fuel structure will facilitate the effective use of TLS data for studying fuel structure dynamics in jack pine stands.

2. Methods

2.1. Modelling objectives

We developed a simulation framework to systematically assess the relative performance of fuel metrics at distinguishing between a wide range of structural conditions. To this end, we created an object library of trees and understory growth by manually segmenting real-world TLS scans obtained in jack pine stands in Northwestern Ontario, Canada, using a Leica BLK360 G1 (Leica Geosystems, Heerbrugg, Switzerland) scanner. This library of forest objects was then used to populate synthetic forest stands based on different structural scenarios represented by a set of four user-modifiable parameters. The synthetic forest stands were used in turn to identify a subset of fuel structure metrics that was best able to distinguish stands with different fuel characteristics. Ground normalization and removal, point cloud denoising, and voxelization of the real-world TLS data were performed using functions within the *lidR*

package (Roussel et al., 2020; Roussel and Auty, 2025). Subsequent processes in this workflow, such as generating synthetic point clouds and computing metrics, were performed using custom R programming language scripts (R Core Team, 2025).

2.2. Structural scenarios

We generated a range of synthetic forest stands with varied fuel structures using four parameters: stem density, understory density, mean tree height and tree mortality proportion. These parameters were chosen because they are relatively simple to extract from vegetation data and meaningfully capture the main axes of structural variability in a forest stand that are pertinent to fire behaviour.

Empirical TLS data for defining the forest objects and parameter space for simulation were derived from 80 scans of jack pine plots in Northwestern Ontario (Fig. 1). All scans were performed using the TLS sampling protocol in the Next Generation Canadian Forest Fire Danger Rating System fuel sampling protocol (NG-CFFDRS) (Boucher et al., 2022, 2024). The NG-CFFDRS TLS protocol is a multi-scan protocol, with one scan at plot center, four scans twelve metres along the cardinal directions and four scans seven metres along the intercardinal directions, with known reference points used to merge the nine scans into one final point cloud (Boucher et al., 2022; Jourdan et al., 2026, Fig. S-1). The jack pine stands scanned were characterised by a relatively open understory and the resulting point clouds exhibit little occlusion. Of the 80 plots scanned, 48 were also sampled using the NG-CFFDRS vegetation sampling protocol (Boucher et al., 2024). These vegetation data

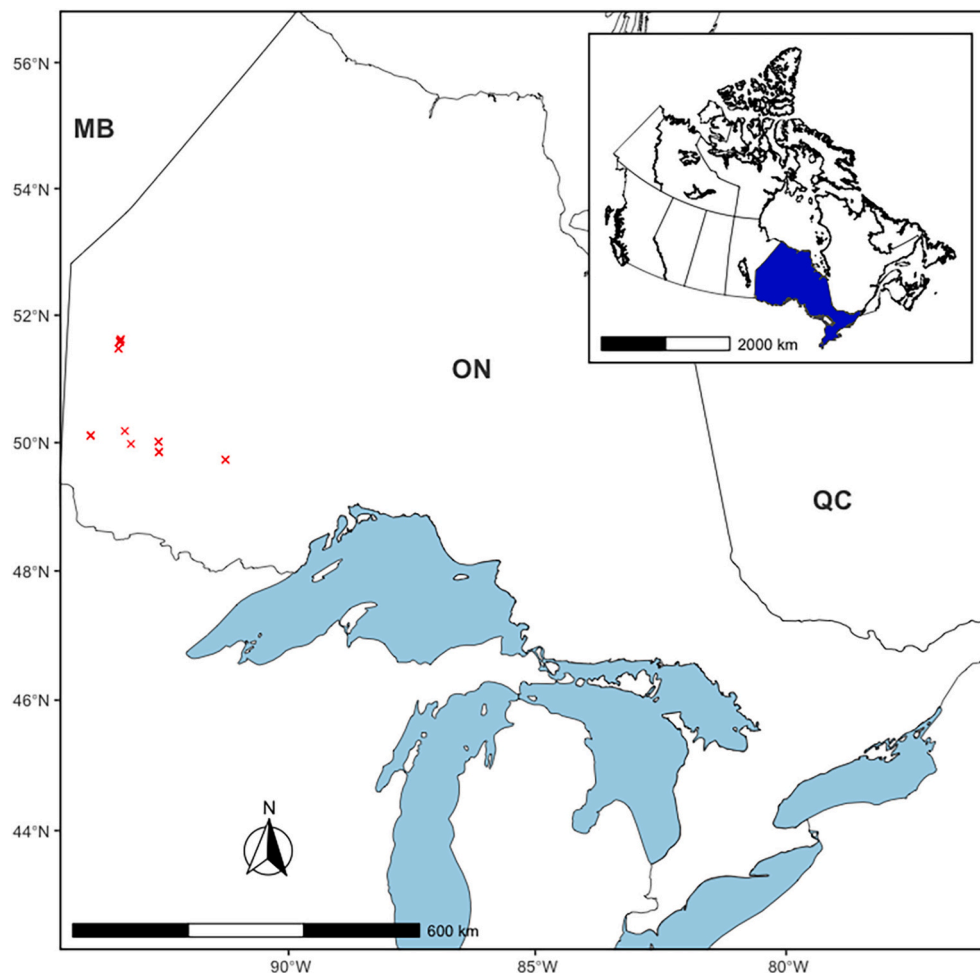


Fig. 1. Location of the 80 jack pine stand scans in Northwestern Ontario, Canada, from which were derived forest objects and parameter space for simulating TLS data. Many scans were done in close proximity (< 1 km apart) and are indistinguishable at the scale of this map.

were used to determine the range of plausible parameter values for each plot (summarised in Table 1). A parameter space to sample from was created by giving 15 possible values for each parameter, spread equally and ranging from the lowest to the highest value observed in real-world data (creating $15^4 = 50,625$ possible parameter combinations). The step size for that parameter was the distance between each consecutive value.

To avoid parameter combinations that were not represented in our real-world data, we fit an initial convex hull around the 4D parameter space values for each plot using the *cxhull* R package (Laurent, 2023). To ensure that the extrema of each parameter were represented in the final parameter combinations, we added a buffer to this initial convex hull. To do this, we took each vertex of the initial convex hull and systematically created eight additional points surrounding it. Specifically, for each parameter, we created two new points by adding and subtracting half the step size specific to that parameter while keeping the other parameter values unchanged. This resulted in eight new points surrounding each vertex, effectively preserving the structure of the convex hull while expanding it to ensure the inclusion of extrema. We then fit a final convex hull around these new points and retained all parameter combinations inside of it (Fig. 2). In total, 7587 parameter combinations remained within the final convex hull and were used to generate the synthetic forest stands.

Each object in a generated scenario was represented by a real tree or shrub segmented from real-world TLS point clouds (Fig. 3). The objects were manually segmented using the 3D point cloud processing software *CloudCompare* (CloudCompare Team, 2022). To reduce computation time, all the objects were voxelized with 5 cm voxels. To select trees for the object library, we chose alive trees and dead stems whose heights spanned the range in the mean tree height parameter. The object library included twenty alive trees, ten dead stems and three understory objects (shrubs or very young tree regrowth).

Synthetic forest stands were generated as circular plots with a 27-m radius, providing a 2-m buffer around the final 25-m radius used for analysis (Fig. 4), which corresponds to the plot size of the real-world TLS data. The location of individual objects (e.g., tree stems, shrubs) within the circular plot was determined using Poisson disk sampling. This method generates random points, typically in a square plot, in a way that avoids the undesirable crowding that can result from getting each object's location at random (Cohen et al., 2003). A Poisson disk sampling algorithm will instead force some distance between the points resulting in a more realistic spread of objects in the stand (Lagae, 2009). The Poisson disk sampling algorithm is very computationally efficient (Bridson, 2007), and is often used for forest simulation to avoid the use of a more complex stand growth method, while maintaining realistic object placement (Ma et al., 2025; Zamuda and Brest, 2013). Here, our goal was to generate plausible patterns of forest stand structure in the absence of explicit pattern-generating processes (Gardner et al., 1987). The algorithm first created a set of “kept points” and added one random point to it. Then, it continuously generated a large set of “possible points” at random, retaining the one that was the furthest from all other points in the set of kept points until the desired number of kept points was met (Mitchell, 1991). To sample the points in a disk instead of a rectangle, we adapted the code from the *PRACMA* R package (Borchers,

Table 1

Range of values for each parameter from which the synthetic point clouds are built. Upper and lower bounds to the ranges were identified from empirical measurements of real-world jack pine stands. Each parameter has fifteen equally spaced possible values.

Parameter	Lowest	Highest
Stem density (# stems/m ²)	0.03	0.2
Understory density (# understory plants/m ²)	0	0.78
Mean tree height (m)	6.54	17.64
Mortality (%)	0	66

Table 2

Stand parameters for example simulated stands shown in Fig. 4.

Scenario	Stem density (# stems/m ²)	Understory density (# understory plants/m ²)	Mean tree height (m)	Mortality (%)
A	0.09	0.28	15.26	56
B	0.08	0.06	12.09	9
C	0.07	0.22	12.88	28
D	0.10	0.00	14.47	14

2023): points were sampled in a square, but those lying outside the disk inscribed in the square were omitted from the “possible points” set. This inscribed disk represented the circular plots. After generating stem locations, a similar method was used to generate the location of the understory objects. These objects were distinct from overstory trees and represented young regrowth and bushes. To prevent understory objects “colliding” with stems, we further adapted the *PRACMA* algorithm by omitting “possible understory points” that were 1 m or less away from the chosen stem points.

Stem heights were sampled from the object library's tree heights using a normal distribution with standard deviation 4.4 and with a mean determined by the scenario's chosen mean tree height parameter. This distribution was chosen because it was the best fit to the observed tree height distribution in the manual measurements. Once the stem locations and heights were determined, they were marked as dead or alive at random, using the mortality proportion as probability for being dead. To avoid biases in metric outputs caused by a consistent orientation of the objects, all objects were assigned a rotation angle drawn uniformly at random, such that all angles were equally likely.

2.3. Generating point clouds from structural scenarios

Once the initial layout was defined, a separate algorithm was used to generate the point cloud for further analysis using segmented objects from the object library.

For each object in the scenario, the algorithm created a copy of its representative object, which was then rotated about the Z-axis, then translated, such that its angle and location matched the scenario parameters. Then, all these objects were combined in a large point cloud, which was cropped to a 25-m radius disk to produce the final generated point cloud for the scenario (Fig. 5).

2.4. Metrics computation

After generating the point cloud, we calculated the twelve candidate metrics (Table 3). Some of these metrics were originally developed for Plant Area Density data (e.g., the CR and ENL metrics in Aalto et al., 2023) or relied on additional preprocessing steps (e.g., distinguishing between occluded and filled voxels in Ehbrecht et al. (2016)). In contrast, our implementations used a simple binary voxel representation of the point cloud (filled or empty). We also made minor modifications to improve computational efficiency, including parallel processing and the use of optimized functions.

UH, FSG, CBH and CH were calculated using the methods outlined in Wilson et al., 2021, with a voxel size of 25 cm. For every column of voxels, FSG is calculated as the length of the longest run of empty voxels. UH is the height of the voxel underneath the longest run of empty voxel. CBH is the height of the voxel above the longest run of empty voxels. CH is the height of the top-most voxel. A stand's final value for these four metrics is the median of these four metrics across all columns (Wilson et al., 2021).

The canopy ratio (CR) metric was adapted from Maeda et al. (2022). We first calculated the 25th and 98th quantile of voxel heights in each column, respectively RH25c and RH98c, for each column of voxels (at 1 m resolution). The stand's overall RH25 and RH98 is the mean of RH25c and RH98c across all columns. The stand's CR is calculated as:

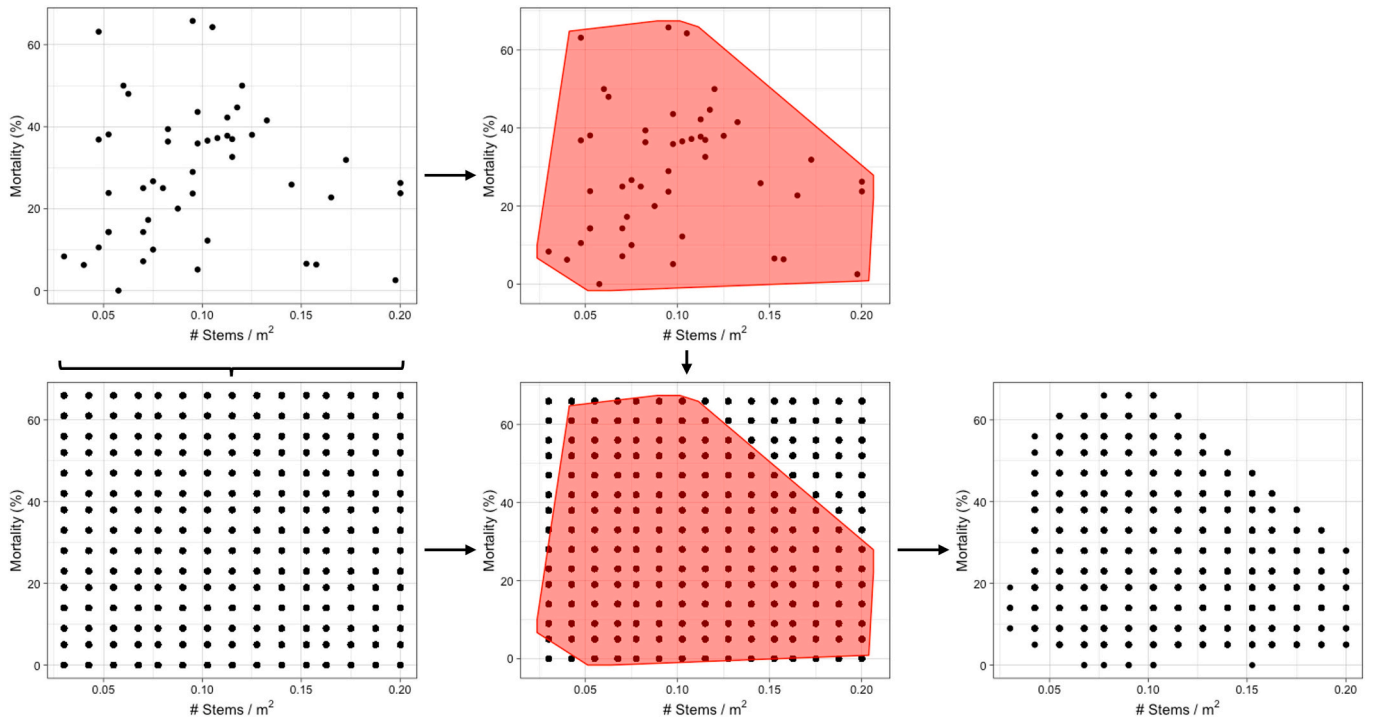


Fig. 2. 2D illustration of the 4D parameter combination selection: values from real-world plot (top left) inform the spread of possible values in the parameter space (bottom left) and a convex hull is drawn around those values (top middle) and used to restrict the final parameter space to be representative of real-world conditions (bottom right).

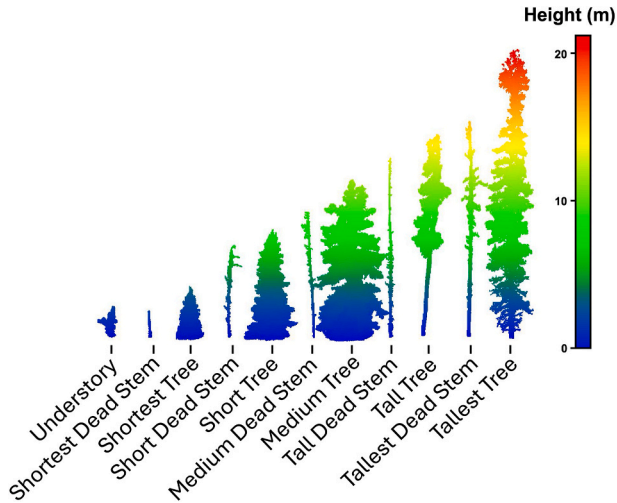


Fig. 3. Eleven of the thirty-three objects included in the object library. Objects were segmented using CloudCompare from Terrestrial lidar (TLS) data taken in Northwestern Ontario. The objects are copied and combined to create synthetic TLS point clouds of a forest stand.

$$CR = \frac{RH98 - RH25}{RH98} \quad (1)$$

Effective Number of Layers (1D ENL, 2D ENL, 2D ENL CV, scan-wide 1D ENL, scan-wide 2D ENL) is a series of metrics that summarizes the number of horizontal layers through the stand. This metric has been used previously to model increased structural complexity resulting from insect outbreaks (Atkins et al., 2020). The different Hill-Numbers (1D and 2D) represent different ways of weighting the layers depending on their density. Horizontal layers are quantified following Aalto et al., 2023's method, which is based on Ehbrecht et al., 2016: The point cloud is first voxelized at a resolution of 20 cm, then divided into 1 m

horizontal slices. In each 1 m $\times$ 1 m column, the number of voxels within each slice is counted and gives the following 1D and 2D ENL values:

$${}^1D \text{ ENL} = \exp \left(- \sum_{i=0}^{top} p_i \ln(p_i) \right) \quad (2)$$

$${}^2D \text{ ENL} = 1 / \sum_{i=0}^{top} p_i^2 \quad (3)$$

where top is the maximum height of the column, and p_i is the number of voxels in each 1 m layer divided by the total number of voxels in the column.

The 1D ENL and 2D ENL value is the mean of their respective ENLs at each column. The 2D ENL CV is the Coefficient of Variation of the 2D ENL values across the columns and indicates the variability of the number of layers throughout the stand. In addition to these column-based ENL values, as in Aalto et al., 2023, we repeat these calculations for the entire scan, without separating the voxels into columns.

Lacunarity is a method to assess spatial heterogeneity across different scales. For each considered scale, it measures the translational similarity of different parts of the data (Plotnick et al., 1993; Saunders et al., 2005). Lacunarity has been used on satellite imagery to model spatio-temporal distributions of forest structure (Malhi and Román-Cuesta, 2008). It has also been used on 2D continuous canopy height data, such as that generated by airborne lidar, to capture cross-scale spatial heterogeneity of forest canopies (Frazer et al., 2005). Cooper, 2022 used a metric derived from lacunarity analysis to assess pre- and post-disturbance differences in forest structure in TLS scans.

Lacunarity is calculated by shifting an even-sided box of varying sizes through the voxelized point cloud. The box's size, r , varies from the size of a voxel to that of the entire point cloud (r_{max}). At each box size, we compute its Lacunarity $L(r)$ as follows:

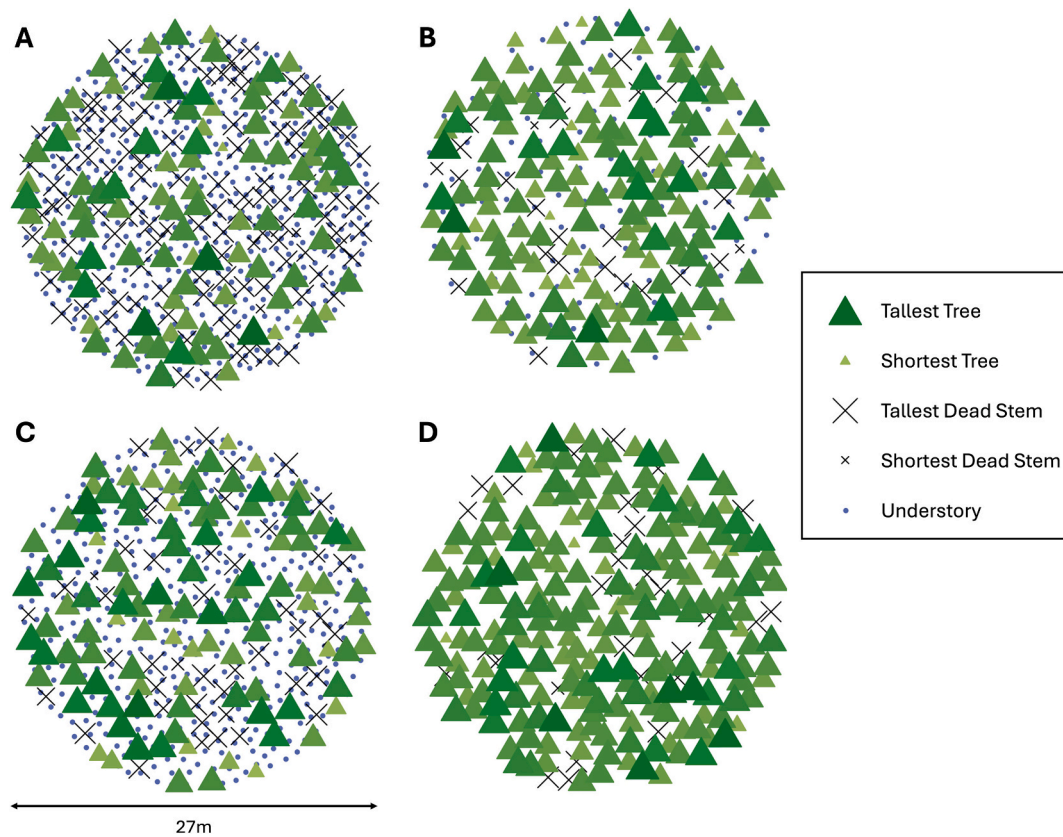


Fig. 4. Representation of four simulated stand layouts. Simulated stands are generated using four parameters each with fifteen levels (See Table 2 for the parameters of these example scenarios). In total, 7587 scenarios were generated, which were then used to create synthetic TLS point clouds.

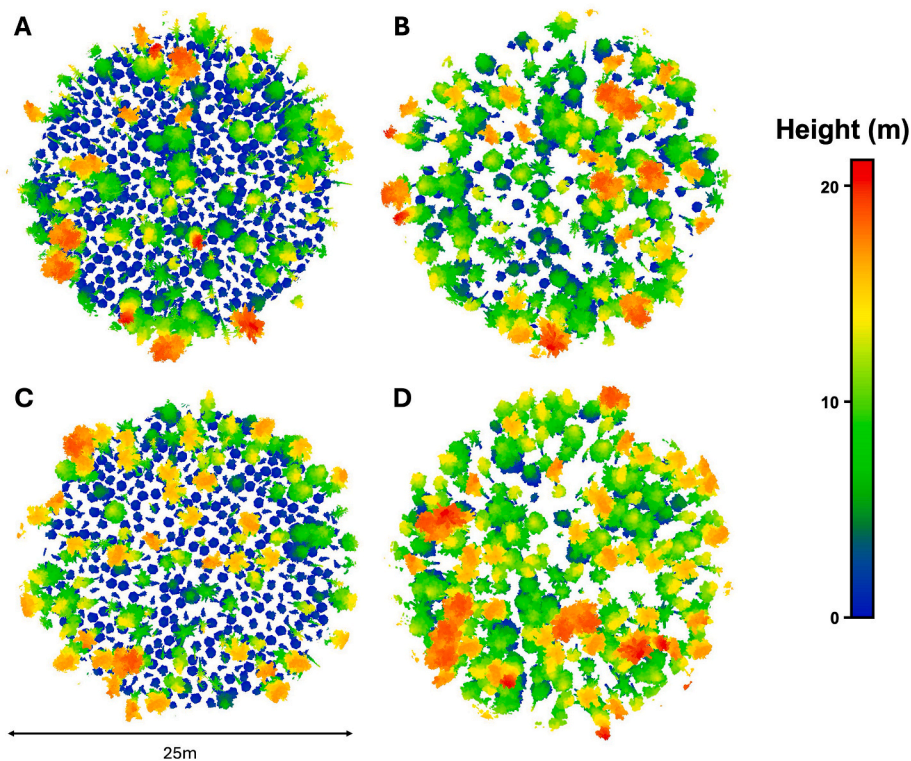


Fig. 5. Top-down render of synthetic TLS point clouds created from four stand scenarios (See Fig. 4 and Table 2 for details about these example scenarios). Each point is coloured based on height from ground. In total, 7587 synthetic TLS point clouds were created.

Table 3

Summary of the 12 metrics tested for their effectiveness in determining fuel structure in jack pine stands. The metrics are classified as: tangible, for those that are easy to understand and recreate classic field measurements; and conceptual, for the novel metrics that can only be obtained with lidar data and are harder to interpret.

	Metric name	Description	Reference
Tangible	Canopy Height (CH)	Height of the canopy at a plot scale	Wilson et al., 2021
	Canopy Base Height (CBH)	Height of the base of the canopy at a plot scale	Wilson et al., 2021
	Fuel Strata Gap (FSG)	Vertical distance between the understory and the canopy at a plot scale	Wilson et al., 2021
	Understory Height (UH)	Height of the understory at a plot scale	Wilson et al., 2021
	Canopy Ratio (CR)	A measure of the vertical concentration of volume in the stand	Maeda et al., 2022
	¹ D Effective Number of Layers (¹ D ENL)	Count of horizontal layers in columns, weighed by the layers' densities	Aalto et al., 2023
	² D ENL		Aalto et al., 2023
Conceptual	Scan-wide ¹ D ENL		Ehbrecht et al., 2016
	Scan-wide ² D ENL		Ehbrecht et al., 2016
	² D ENL CV	Coefficient of variation of the ² D ENL metric across columns	Aalto et al., 2023
	Total Lacunarity	Indices summarizing a stand's lacunarity curve, which represents structural heterogeneity at different scales	Cooper, 2022
	Heterogeneity Index		Scott et al., 2022

$$L(r) = \sum_{r=1}^{r_{\max}} M^2 \times Q(M, r) / \left(\sum_{r=1}^{r_{\max}} M \times Q(M, r) \right)^2 \quad (4)$$

where M is the mass of the box: the number of occupied voxels in it, and $Q(M, r)$ is the frequency of that mass for that box size, calculated by dividing the number of boxes with that mass with the total number of boxes. The output of lacunarity analysis can be represented as a curve showing spatial heterogeneity, $L(r)$, as a function of scale, r .

We summarised the lacunarity using two methods: Total Lacunarity (Cooper, 2022) and Scott's Heterogeneity Index (Scott et al., 2022). They are calculated as follows by summing over all possible box sizes from 1 to $r_{\max}$:

$$\text{Total Lacunarity} = \frac{1}{L(1)} \times \sum_{r=1}^{r_{\max}} L(r) L(r) \quad (5)$$

$$\text{Heterogeneity Index} = 1 - \frac{2}{(L(1) \times r_{\max})(1 + r_{\max})} \times \sum_{r=1}^{r_{\max}} r L(r) \quad (6)$$

Following Cooper (2022), we voxelized the point cloud with a resolution of 50 cm.

2.5. Analysis

Our goal was to find the smallest subset of metrics that can arrange the synthetic stands into different structural groups. In other words, we aimed to identify a parsimonious subset of metrics that optimally split the synthetic jack pine structural scenarios into distinct clusters. We implemented a two-step metric selection strategy: first, identifying and removing redundant metrics. Then, identifying which of the non-redundant metrics can be used to separate the stand scenarios into clusters.

The first step of the metric reduction strategy is to make a subset with

all non-redundant metrics. We used the *findCorrelation* function from the *caret* R package (Kuhn, 2008) which identifies then removes in step the metrics with the highest mean absolute correlation. We tested different Pearson correlation thresholds, which determine the number of metrics retained: none of the pairs of metrics in the resulting subset have a Pearson correlation above that threshold. We explored the sensitivity to different thresholds and picked one where the number of retained metrics stabilizes.

To further explore the non-redundant subset of metrics, we used a Multivariate Regression Tree (MRT) (De'ath, 2002) as implemented in the *mvpart* R package (De'ath, 2007). The MRT forms clusters of synthetic stands based on repeated splitting, where each split is determined by the value of a single metric chosen to minimize the dissimilarity of the scenario parameters within clusters. Here, the response was the scenario parameters, and the predictors were the metrics.

With increasing splits in the MRT, the relative error of the tree decreases. To determine the parsimonious tree size, we used the procedure outlined by De'ath, 2002 and looked at the cross-validated relative error (CVRE) at each size, choosing the smallest tree whose CVRE is within one standard-deviation of the smallest CVRE.

We linearly standardized the parameters before running the MRT, and the resulting tree has splits based on the value of a subset of the non-redundant fuel metrics. This is considered the parsimonious subset whose combination is best able to separate the structural scenarios.

To evaluate the performance of the non-redundant metrics, and of the final parsimonious subset, at explaining the variance in structural parameters, we used a redundancy analysis (RDA) (Legendre and Legendre, 2012), as implemented in the *rda* function of the *vegan* R package (Oksanen et al., 2022). RDA is a constrained ordination technique which can show how much of a response matrix is explained by one or several predictor vectors (Legendre and Legendre, 2012). In this case, because we're interested in how well the metrics predict actual structural conditions, the response data are the matrix of structural parameters, and the predictor data are the fuel metrics. This procedure allowed us to evaluate how much explanatory power was lost when using the parsimonious subset instead of the non-redundant metrics subset by comparing the adjusted R^2 of their respective redundancy analysis model.

To validate our simulation data framework, we compared the metric outputs of 48 real-world TLS scans, for which we also have manual vegetation measurements, with their synthetic stand equivalents. For each real-world TLS scan, we generated 50 synthetic stands with the same structural parameters. We then calculated the z-score, which gives the number of standard deviation units of a metric compared to the mean of the synthetic plots (Legendre and Legendre, 2012). Because the parameters are a simplified representation of the real-world stands, we anticipated that the metric outputs of the real-world stand would differ from that of their synthetic equivalents but that the difference would remain consistent across the real-world stands.

3. Results

3.1. Non-redundant metrics

We computed the output of the twelve candidate metrics on the 7587 synthetic TLS point clouds, and explored which ones were redundant by removing the metrics with the highest mean absolute correlation, until all metrics were under our chosen Pearson correlation threshold. We selected a Pearson correlation threshold of 0.8, because it represented a balanced trade-off between aggressively removing metrics and maintaining valuable information (Supplementary Fig. S-2).

Using the process, we identified six metrics (Fuel Strata Gap, Understory Height, Canopy Ratio, Coefficient of Variation of ²D Effective Number of Layers CV, Scan-wide ¹D Effective Number of Layers, and Total Lacunarity) to retain (Fig. 6). Based on the RDA, these six metrics explain 78.9% of the variation in structural parameters (Table 4).

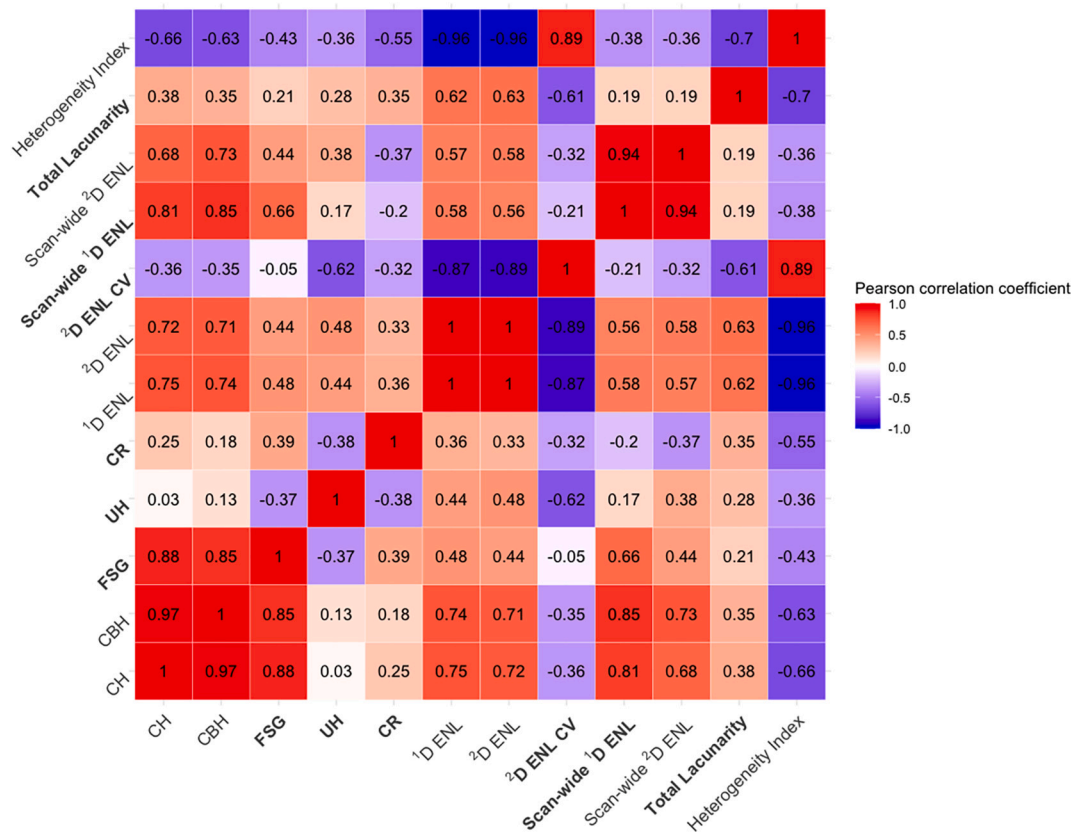


Fig. 6. Correlation heatmap of the twelve metrics. After the removal of highly correlated metrics, only six metrics (FSG, UH, CR, ²D ENL CV, Scan-wide ¹D ENL and Total Lacunarity) remain.

Table 4

ANOVA of the redundancy analysis (RDA) of scenario parameters as explained by the six non-redundant metrics (FSG, UH, CR, ²D ENL CV, Scan-wide ¹D ENL, Total Lacunarity). The six non-redundant metrics explain 78.9% of the variance in scenario parameters.

	Degrees of freedom	Variance	Proportion explained	F-value	P-value
Model	6	83,759	0.789	4721.5	< 0.001
Residual	7580	22,412	0.211		

3.2. Metrics ranking

The non-redundant metrics were further reduced by doing a Multivariate Regression Tree, where the structural scenarios were split into clusters based on the values of the metrics. The splits in the multivariate regression tree were based on four (FSG, CR, ²D ENL CV, and Scan-wide ¹D ENL) of the six non-redundant predictor metrics, and the scenarios were split into 11 clusters (Fig. 7, Supplementary Fig. S-3). In an RDA, these four metrics explain 78.0% of the variation in structural parameters (Table 5).

3.3. Comparison with empirical TLS data

From the difference of metric outputs between real-world stand TLS data and the synthetic stands that have the closest structural parameters (Fig. 8), our simulation method systematically underestimates ²D ENL CV and FSG and overestimates UH. However, it is consistent for CR, Scan-wide ¹D ENL and Total Lacunarity.

4. Discussion

In this study, we developed and presented a novel framework for generating synthetic forest stands from TLS-derived point cloud objects and using these stands to compare TLS metrics. Considering the many metrics defined in the literature, it is important to evaluate their respective performance when selecting a set that can reliably characterize and distinguish among different fuel structures. In total, we simulated 7587 stands of varying structural conditions and tested twelve metrics. To ensure realistic parameter combinations were used, the synthetic stands were generated by sampling within an area of the parameter space constrained by a convex hull of attributes of real-world stands. Six of the twelve metrics were retained after a correlation analysis (Fuel Strata Gap, Understory Height, Canopy Ratio, Coefficient of Variation of ²D Effective Number of Layers CV, Scan-wide ¹D Effective Number of Layers, and Total Lacunarity). Of those six metrics, four (Fuel Strata Gap, Canopy Ratio, Coefficient of Variation of ²D Effective Number of Layers CV and Scan-wide ¹D Effective Number of Layers) were the basis for splitting the 7587 simulated stands into 11 structurally distinct clusters. The four metrics identified through the MRT process performed similarly to the six non-redundant metrics (78.0% vs 78.9% of the variation in parameters explained). Therefore, the omission of the Total Lacunarity and Understory Height metrics from the final metric subset was not impactful.

Despite their simplicity, our chosen parameters are effective at capturing a broad range of structural scenarios relevant to fires. For example, low tree height and high understory density can be associated with higher crowning potential due to the likely presence of ladder fuels when compared with a stand with very little understory density and very tall trees. Similarly, a high stem density will be associated with crown connectivity and crown fire sustainability, whereas low stem density or high stem mortality would imply a lesser chance that a crown fire would

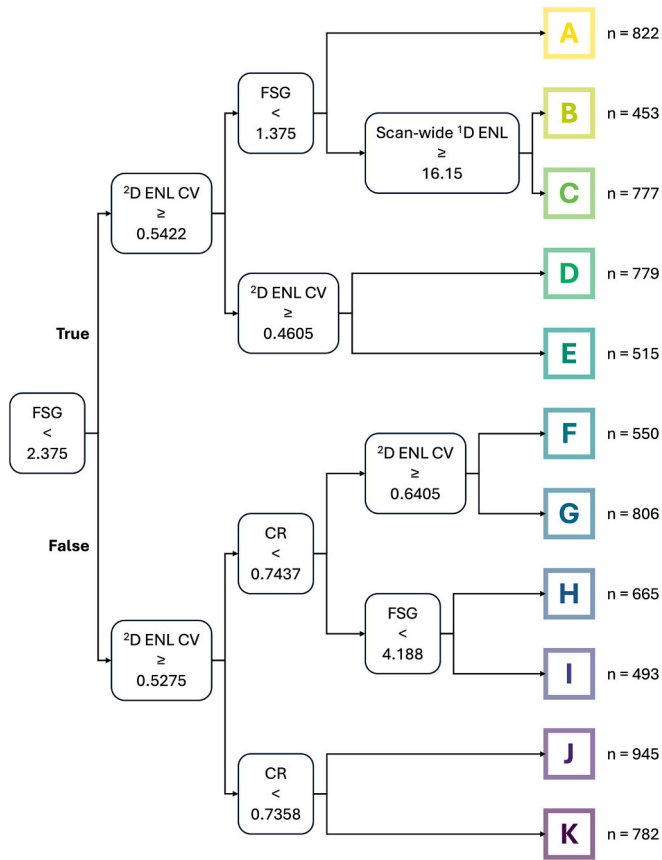


Fig. 7. Multivariate Regression Tree (MRT) splitting synthetic TLS data in 11 clusters based on the value of four metrics (FSG, Scan-wide ¹D ENL, ²D ENL CV and CR). Each of the 11 clusters identified by the MRT is represented by a letter. The clusters' mean parameter values can be found in Fig. S-3.

Table 5

ANOVA of the redundancy analysis (RDA) of scenario parameters as explained by the four metrics determining the splits in the Multivariate Regression Tree (FSG, Scan-wide ¹D ENL, CR, ²D ENL CV). The four metrics explain 78% of the variance in scenario parameters.

	Degrees of freedom	Variance	Proportion explained	F-Value	P-Value
Model	4	82,762	0.780	6701.7	<0.001
Residual	7582	23,408	0.221		

be sustained.

Beyond identifying a parsimonious subset of TLS-derived fuel metrics for jack pine stands, the workflow presented here provides a general framework for evaluating and interpreting structural metrics under controlled conditions. Although we began with an a priori subset of twelve metrics representing a range of approaches, the framework itself places no restriction on the number or type of metrics evaluated. Additional metrics could readily be incorporated and ranked according to their ability to distinguish structural scenarios. Likewise, the initial correlation filtering step was included to reduce redundancy and simplify interpretation but is not required. In applications where predictive performance is prioritized, the clustering procedure alone could instead be used to identify the most informative subset without any initial filter.

The synthetic point cloud framework can also be refined. The parameters chosen to simulate the forest stands, and the mechanisms underlying the point cloud generation, do not completely reflect the complex mechanisms driving forest stand growth. For example, a tree in

a dense stand will not grow in the same way as a tree in a sparse stand, which affects the vertical and horizontal fuel distribution in stands of different density (Jucker et al., 2015). Further, tree and shrub locations in real-world stands are driven by complex mechanisms, such as wind dispersal, solar radiation, individual- and species- specific attributes and site-specific disturbance history (Kenkel et al., 1997; North et al., 2004; Smirnova et al., 2008). In addition, the trees in the object library were only selected based on their height, without a systematic consideration for their surrounding density. While this selection process ensured that the tree heights span the observed range, it might not replicate the full range of variation in tree structure. Lastly, the systematic overestimation of Fuel Strata Gap and underestimation of Understory Height compared to real-world stands suggest that the current object library does not fully represent variation in jack pine understory vegetation. Expanding the diversity and architecture of understory objects would likely improve the realism of simulated stands and reduce these biases. However, the consistent direction in the bias of those metrics suggests that their overall trends are captured by the synthetic point cloud framework proposed.

Nevertheless, this accessible and empirically based method of generating synthetic forest stands exhibits several desirable qualities that enable the creation of a wide range of stand types. First, its relatively simple set of parameters reduces the complexity of the process. Second, the method for placing objects within synthetic stands is straightforward and requires little prior knowledge of a forest's expected growing patterns. A natural future improvement would be to refine and expand the object library, with a particular focus on understory objects, which could be achieved with the aid of automatic segmentation algorithms, for example with Xi and Hopkinson, 2022's *Treeiso*. Our approach could further be refined to improve realism in object placement by using more advanced techniques, such as implementing variable radii Poisson Disk Sampling (Yan and Wonka, 2013) or using empirical data to develop a generative model that reproduces observed spatial patterns and guides the placement of synthetic objects (Hutchings et al., 2024). Additional refinement of the synthetic point cloud generation framework may alter which metrics are ultimately retained. Such improvements must, however, be balanced against computational tractability. For example, adding an additional stand parameter to the simulations (e.g. mean understory height) while retaining the fifteen possible values would increase the number of unique parameter combinations from 50,625 to 759,375. Future development of the framework will therefore require balancing ecological realism with computational efficiency, while retaining the ability to generate sufficiently diverse structural scenarios for robust metric evaluation.

While the parameters chosen to describe the synthetic stands perform well for the objectives of this study, they could be modified or replaced for studying other forest attributes beyond fuel structure. Likewise, the object library could be modified to test metrics in other forest types.

5. Conclusion

This research adds to the growing body of work using lidar to understand fuels in forested areas. TLS-derived fuel characteristics are an excellent tool for precise fuel modelling, but the data are complex, and it is challenging to determine in what ways they should be summarised. While lidar enables the collection of precise structural data, it remains surprisingly difficult to translate its abundance of information into meaningful insights (Abdollahi and Yebra, 2025; Stefanidou et al., 2020; Xi et al., 2023), highlighting a gap between data availability and actionable understanding. Because of this, we recommend that researchers and practitioners carefully test the metrics used to ensure that they are interpreted correctly. Because real-world data are often limited, synthetic data provide a valuable alternative or complement for evaluating the analytical tools used to characterize and infer fuel structure

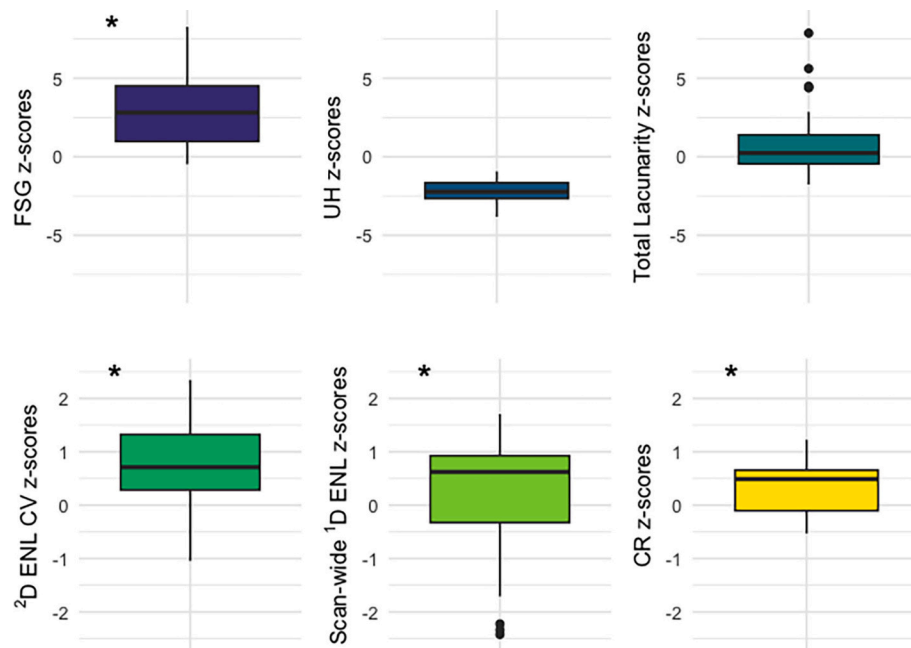


Fig. 8. Boxplots summarizing 48 z-scores (one per real-world stand), where each compares a real-world terrestrial lidar stand's metric value to the mean metric value of 50 synthetic stands generated with the same structural parameters. For the MRT-identified structural metrics (indicated by an asterisk), 2D ENL CV, Scan-wide 1D ENL and CR of real-world TLS data are usually within one standard deviation larger than their synthetic equivalents. However, FSG is systematically between one to five standard deviations larger than synthetic equivalents. For the other two non-redundant metrics, Total Lacunarity is consistent with synthetic stands, while UH is systematically between two to three standard deviations smaller.

from lidar across the full range of possible conditions. Synthetic point clouds enable the assessment of metric behaviour across forest structural conditions that are often underrepresented in empirical datasets. Building on previous applications of synthetic spatial data for metric evaluation (e.g. Fortin et al., 2003), we demonstrate that TLS-derived fuel metrics can be systematically compared and better understood using a simple synthetic point cloud generation framework.

By testing existing fuel metrics in a jack pine stand context, this study promotes the use of lidar for understanding fuels in the boreal forest of Northwestern Ontario. The synthetic data protocol described in this study can also be reused for studies in other forest types by modifying the parameters used for generating the synthetic data, to capture a range of scenarios of interest and use those scenarios to test metrics and other algorithms. Generating synthetic data to test metrics allows the exploration of different structural scenarios and helps interpret the metrics that provide more abstract information. Effective interpretation of these abstract metrics will enable future research to take advantage of all the complexity of lidar data without restricting analyses to only metrics replicating traditional measurements.

Software and data availability

All Scripts and input data used in this study can be found on Github at https://github.com/JamesLabUofT/Synthetic_TLS_Metric_Selection

CRedit authorship contribution statement

Léo Jourdan: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Patrick M.A. James:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecoinf.2026.103977>.

Data availability

Data used is linked in the manuscript

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